

Lezione 1 - Studi di funzione

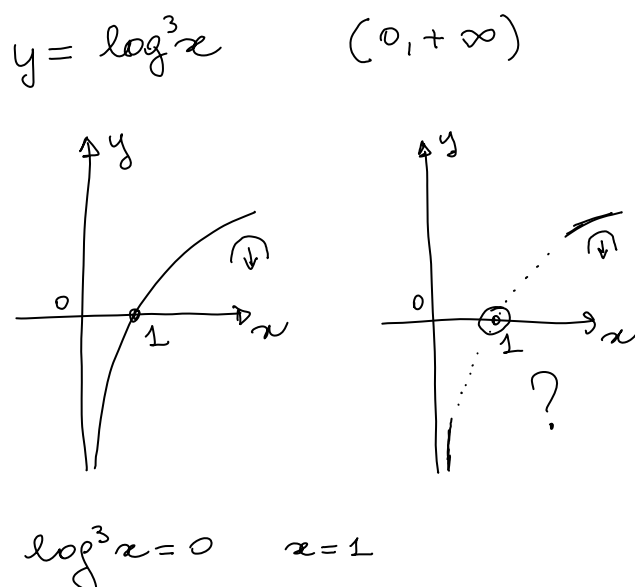
Matematica del continuo

Modulo 11 - Studio di funzioni

Unità didattica 2 – Funzioni con logaritmi ed esponenziali

Enrico Spoletini

Università degli Studi di Milano - SSRI - CdL Online



$$y = \log^3 x$$

$$y' = 3 \log^2 x \cdot \frac{1}{x}$$

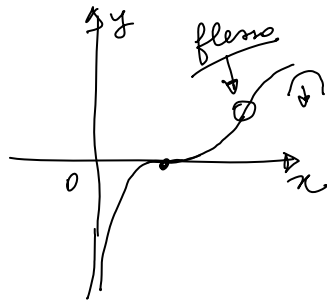
$$y' \geq 0 \quad \text{sempre}$$

$$y' = 0 \quad x = 1$$

↑

Tg orizz

y è sempre cresc



$$y' = 3 \log^2 x \cdot \frac{1}{x}$$

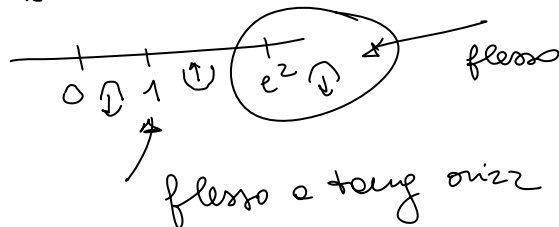
$$y'' = 3 \left(2 \log x \cdot \frac{1}{x} \cdot \frac{1}{x} + \log^2 x \cdot \left(-\frac{1}{x^2} \right) \right)$$

$$= \frac{3 \cdot \log x}{x^2} (2 - \log x)$$

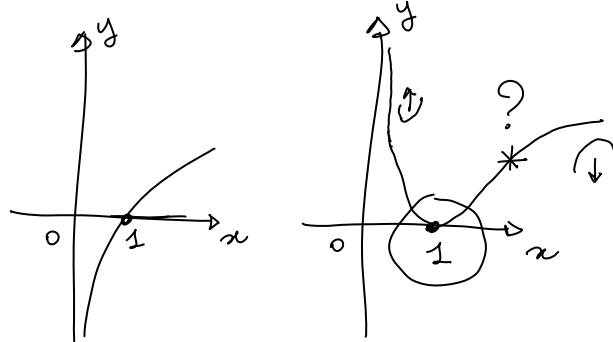
$$y'' = 0 \quad \log x = 0 \quad \log x = 2$$

$$x = 1 \quad x = e^2$$

$$y'' > 0$$



$$y = \log^4 x$$



$$\mathbb{R} \quad (0, +\infty) \quad \lim_{x \rightarrow 0^+} y = +\infty \quad \lim_{x \rightarrow +\infty} y = +\infty$$

$$y = \log^4 x$$

$$y' = 4 \log^3 x \cdot \frac{1}{x}$$

$x < 1$ $x > 1$ $x = 1$
 $y' < 0$ $y' > 0$ minimum

$$y'' = 4 \left(3 \log^2 x \cdot \frac{1}{x^2} - \log^3 x \cdot \frac{1}{x^2} \right)$$

$$= \frac{4 \log^2 x}{x^2} \underbrace{(3 - \log x)}$$

$\uparrow x = e^3$
flessa

$$y'' = 0$$

